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Molecular dynamics simulation to reveal the transport mechanism of LiPF_6 in ethylene carbonate + dimethylcarbonate binary solvent ✓

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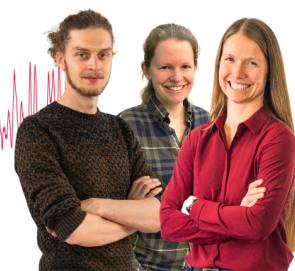


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ABSTRACT

This paper presents the molecular dynamics simulation of 1 mol kg^{-1} LiPF_6 in a binary solvent of ethylene carbonate (EC) and dimethylcarbonate, which is a representative electrolyte solution for lithium-ion batteries. The simulation successfully reproduced the diffusion coefficient, ionic conductivity, and shear viscosity as functions of EC content at 300 K, which had been experimentally determined in our previous study. The Yukawa potential was adopted to model interchange interactions to reduce computational costs, which consequently allowed us to precisely calculate the conductivity and viscosity by directly integrating time-correlation functions without explicitly modeling the molecular polarization. Breaking down microscopic current correlation functions into components revealed that, whereas the cation–anion attractive interaction dominantly impedes the conduction when the EC content is low, it is the cation–cation and anion–anion repulsive interactions that reduce the conductivity at a high EC content. An analysis of the pressure correlations revealed that all components positively contribute to the viscosity in the binary solvent without the electrolyte. On the other hand, negative terms are observed in five out of six cross correlations in the presence of the electrolyte, implying that these correlations negatively contribute to the shear stress and entropy production, both of which are net positive.

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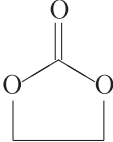
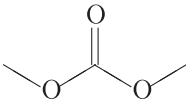
I. INTRODUCTION

Nonaqueous electrolyte solutions provide an important matrix through which charge carriers are transported between the electrodes of lithium-ion batteries (LIBs). The transport phenomena (e.g., diffusivity, ionic conductivity, and viscosity) at an electrolyte concentration at which these electrochemical devices function ($\sim 1 \text{ mol kg}^{-1}$) are considerably influenced by correlations between ions, which are insignificant in dilute solutions. For example, the conductivity of concentrated solutions is often less than that estimated from the diffusivity of the ions, for which the cation–anion correlation, in terms of ion dissociation, is often considered responsible. A solvent with high permittivity (e.g., ethylene carbonate, EC, Table I) is mixed with other solvents with high fluidity (e.g., dimethylcarbonate, DMC, Table I) to facilitate ion dissociation at the expense of the poor fluidity of the former.¹ An epitome would be 1 mol kg^{-1} lithium hexafluorophosphate (LiPF_6) dissolved

in an EC/DMC binary solvent, wherein the specific conductivity initially increases with the increasing EC content, despite the monotonous increase in viscosity [cf. Figs. 6(a) and 7]. Based on the experimentally measured diffusion coefficient, conductivity, viscosity, and Raman spectroscopy, our previous studies^{2–4} revealed the transport mechanism of lithium electrolytes dissolved in binary solvents by referring to concepts such as the preferred Li^+ solvation of EC over DMC, degree of ion dissociation, and static and dynamic radii of ions. However, the discussion has been somewhat circuitous because of the difficulty in directly assessing the dynamic correlations between ions and molecules via experimental means. Theoretical considerations, such as molecular dynamics (MD) simulations, are hence called for.

In this study, based on classical MD simulations, the transport properties of the EC/DMC binary solvent and its 1 mol kg^{-1} LiPF_6 solution were calculated as a function of the EC content by directly integrating current and pressure correlation functions. The

TABLE I. Ethylene carbonate (EC) and dimethylcarbonate (DMC); ϵ_r : dielectric constant; η : shear viscosity (see, for example, Ref. 5 and those cited therein).

		
	Ethylene carbonate (EC)	Dimethylcarbonate (DMC)
ϵ_r	89.6	3.1
$\frac{\eta}{\text{mPa s}}$	1.85(313 K)	0.59(298 K)

simulation successfully reproduced the experimental results² of not only the diffusion coefficient, conductivity, and viscosity but also the solution density and fraction of DMC conformers. Several studies dealt with the MD simulation of the related electrolyte systems,^{6–10} yet many of them focus on the structural aspects (e.g., the structures of the Li^+ -solvation shell and ion pairs), and a few have briefly addressed the transport properties. Compared to structural information, a considerable computational tool is required to obtain dynamic properties from correlation functions because a long simulation run and a number of data points are necessary for their integration to statistically converge. For calculating the ionic conductivity of electrolyte solutions, Borodin and Smith¹¹ contrived a scheme to mitigate this statistical problem based on the integrated form of correlation functions,¹² and they estimated the ratio successfully between the auto-correlation and cross correlation, which corresponds to $1 - \Delta$ in Eq. (43) without directly considering the long time limit of the velocity correlation functions. However, to our knowledge, no study has so far analyzed the correlation functions by accurately breaking them down into components throughout a range of the solvent composition. We substituted the Yukawa (or Debye–Hückel) potential for electrostatic interactions [cf. Eq. (34)] to reduce the computational load. Without explicitly modeling the molecular polarization or many-body interactions, which would be computationally expensive, the Yukawa potential allowed us to satisfactorily integrate correlation functions. We untangled the transport mechanism by breaking down the correlation functions into components that are not directly observable experimentally.

II. THEORETICAL BACKGROUND

A. Correlation function and its breakdown into components

The transport properties considered in this study were derived from the time integral of correlation functions in the framework of the Green–Kubo (GK) formulation based on the linear response theory.^{13–16} For a physical quantity A , the correlation function $y(A; t)$ pertinent to this study has the form

$$y(A; t) \equiv \left\langle \sum_i \sum_j A_i(0) A_j(t) \right\rangle, \quad (1)$$

where $A_i(t)$ and $\langle \cdots \rangle$ are the quantity A of atom i at time t and the ensemble average, respectively. When A is a vector $\mathbf{A}$ or a tensor $\mathbf{A}$, the operation in Eq. (1) is read as its scalar product, $\mathbf{A}_i(0) \cdot \mathbf{A}_j(t)$ or

$A_i(0) : A_j(t)$ [cf. Eq. (24) for the latter]. The transport properties are expressed in terms of the long-term integral $Y(A)$ of $y(A; t)$ as

$$Y(A) \equiv \lim_{\tau \rightarrow \infty} \int_0^\tau y(A; t) dt. \quad (2)$$

If the system comprises molecular species α (e.g., DMC, EC, Li^+ , or PF_6^-), $Y(A)$ can be broken down into the following components:

$$Y(A) = \sum_\alpha \left(Y_{\alpha\alpha}(A) + 2 \sum_{\beta > \alpha} Y_{\alpha\beta}(A) \right), \quad (3)$$

where

$$Y_{\alpha\alpha}(A) \equiv \lim_{\tau \rightarrow \infty} \int_0^\tau \left\langle \sum_{i \in \alpha} \sum_{j \in \alpha} A_i(0) A_j(t) \right\rangle dt, \quad (4)$$

$$Y_{\alpha\beta}(A) \equiv \lim_{\tau \rightarrow \infty} \int_0^\tau \left\langle \sum_{i \in \alpha} \sum_{j \in \beta} A_i(0) A_j(t) \right\rangle dt \quad (\alpha \neq \beta), \quad (5)$$

where the properties of the time correlation function stipulate $Y_{\alpha\beta}(A) = Y_{\beta\alpha}(A)$. Equations (4) and (5) represent the contributions from the correlation between molecules belonging to the same species α and different species α and β (cross correlation between α and β), respectively.

The autocorrelation term

$$Y_\alpha(A) \equiv \lim_{\tau \rightarrow \infty} \int_0^\tau \left\langle \sum_{i \in \alpha} A_i(0) A_i(t) \right\rangle dt \quad (6)$$

is computationally intensive for a direct calculation. However, when it is available, as in conductivity, $Y_{\alpha\alpha}$ can be further divided into an auto-term and a cross term as

$$Y'_{\alpha\alpha} = Y_{\alpha\alpha} - Y_\alpha, \quad (7)$$

where

$$Y'_{\alpha\alpha}(A) \equiv \lim_{\tau \rightarrow \infty} \int_0^\tau \left\langle \sum_{i \in \alpha} \sum_{j \in \alpha} A_i(0) A_j(t) \right\rangle dt \quad (i \neq j), \quad (8)$$

where the prime symbol ' is added to indicate the constraint $i \neq j$ and distinguish it from Eq. (4). Equation (8) represents the cross correlation ($i \neq j$) between molecules belonging to the same species α . In this case, Eq. (3) becomes

$$Y(A) = \sum_\alpha \left(Y_\alpha(A) + Y'_{\alpha\alpha}(A) + 2 \sum_{\beta > \alpha} Y_{\alpha\beta}(A) \right). \quad (9)$$

B. Diffusion coefficient D_α

The Einstein–Smoluchowsky equation relates the self-diffusion coefficient of species α , D_α to its mean square displacement $\langle \Delta \mathbf{r}_i^2 \rangle_{i \in \alpha}(t)$ and time t in the long time limit where molecular motion is considered in the diffusive regime. In a three-dimensional system, this is given as

$$D_\alpha = \lim_{t \rightarrow \infty} \frac{\langle \Delta \mathbf{r}_i^2 \rangle_{i \in \alpha}(t)}{6t}. \quad (10)$$

Alternatively, the GK formulation yields the expression D_α in terms of the integrated velocity autocorrelation. By setting $A = \mathbf{v} = \{\mathbf{v}_i\}$ (=the velocity of atom i) in Eq. (6), we obtain the corresponding expression,

$$D_\alpha = \frac{Y_\alpha(\mathbf{v})}{3N_\alpha}, \quad (11)$$

where N_α is the number of molecules that belong to α . Equations (10) and (11) imply that the integrated velocity autocorrelation $Y_\alpha(\mathbf{v})$ can be obtained from the mean square displacement $\langle \Delta r_i^2 \rangle_{i \in \alpha}(t)$, which is relatively easy to calculate; cf. Eqs. (14) and (15).

C. Ionic conductivity σ

The GK formulation relates the specific conductivity σ to the microscopic current correlation as the limit $\omega \rightarrow 0$ of the frequency-dependent response of the system to an electric field perturbation, where ω is the electric field frequency,

$$\sigma = \frac{Y(q\mathbf{v})}{3kTV}, \quad (12)$$

where $q = \{q_i\}$, k , T , and V are the charge on the atoms, Boltzmann constant, absolute temperature, and system volume, respectively. Equation (9) suggests that the specific conductivity σ can be broken down as

$$\sigma = \sum_\alpha \left(\sigma_\alpha + \sigma'_{\alpha\alpha} + 2 \sum_{\beta > \alpha} \sigma_{\alpha\beta} \right). \quad (13)$$

The autocorrelation term σ_α is related to the diffusion coefficient D_α as

$$\sigma_\alpha \equiv \frac{Y_\alpha(q\mathbf{v})}{3kTV} \quad (14)$$

$$= \frac{(ez_\alpha)^2 Y_\alpha(\mathbf{v})}{3kTV} \\ = \frac{N_\alpha (ez_\alpha)^2 D_\alpha}{kTV} \quad \because \text{Eq. (11)}, \quad (15)$$

where $|e|$ and z_α are the protonic charge and valence of species α , respectively. The autocorrelation term σ_α , often referred to as the Nernst–Einstein conductivity, corresponds to the hypothetical conductivity that would be observed if the diffusion of individual ions were uncorrelated with each other. The cross correlation terms in Eq. (13), $\sigma'_{\alpha\alpha}$ and $\sigma_{\alpha\beta}$, are expressed as

$$\sigma'_{\alpha\alpha} \equiv \frac{Y'_{\alpha\alpha}(q\mathbf{v})}{3kTV} \\ = \frac{(ez_\alpha)^2 Y'_{\alpha\alpha}(\mathbf{v})}{3kTV} \quad (16)$$

$$= \frac{(ez_\alpha)^2 \{Y_{\alpha\alpha}(\mathbf{v}) - Y_\alpha(\mathbf{v})\}}{3kTV} \quad \because \text{Eq. (7)} \\ = \frac{(ez_\alpha)^2 Y_{\alpha\alpha}(\mathbf{v})}{3kTV} - \sigma_\alpha \quad \because \text{Eq. (14)}, \quad (17)$$

$$\sigma_{\alpha\beta} \equiv \frac{Y_{\alpha\beta}(q\mathbf{v})}{3kTV} \\ = \frac{e^2 z_\alpha z_\beta Y_{\alpha\beta}(\mathbf{v})}{3kTV}. \quad (18)$$

These terms represent the contribution of the cross correlation between the same species α and the cross correlation between different species α and β .

D. Shear viscosity η

The pressure tensor $\mathbf{P}$ is expressed as the sum of its atomic components, $\mathbf{P}_i$ ($\mathbf{P} = \sum_i \mathbf{P}_i$), as

$$\mathbf{P}_i V \equiv m_i \mathbf{v}_i \mathbf{v}_i + \mathbf{r}_i \sum_{j \neq i} \mathbf{f}_{i \leftarrow j}, \quad (19)$$

where $\mathbf{f}_{i \leftarrow j}$ is the force acting on the atom i from j . The exterior or ordered product $\mathbf{AB}$ of the two vectors $\mathbf{A}$ and $\mathbf{B}$ (also termed a “dyad”) is defined as

$$(\mathbf{AB})^{ab} \equiv A^a B^b \quad (a, b = x, y, z), \quad (20)$$

where A^a is the a -component of $\mathbf{A}$. The virial theorem suggests that the first term on the right-hand side of Eq. (19) is related to the pressure (i.e., the impulse applied to the wall) of the ideal gas and the second term represents the deviation from the ideal gas caused by the interaction between atoms. Tensor $\mathbf{P}$ can be divided into

$$\mathbf{P} = (p + \pi) \mathbf{U} + \Pi, \quad (21)$$

where scalar p is the pressure observed when the system is in equilibrium, $\mathbf{U}$ is the unit tensor ($U^{ab} = \delta_{ab}$), scalar $-\pi$ is the bulk stress, and traceless tensor $-\Pi$ ($\Pi^{ab} \delta_{ab} = 0$) is the shear stress. The GK formalism suggests that Π can be calculated by integrating the correlation of VII or Π as

$$\eta = \frac{Y(\text{VII})}{6kTV} \quad (22)$$

$$= \frac{VY(\Pi)}{6kT}, \quad (23)$$

where the operation in Eq. (1) is read as the scalar product of the two tensors,

$$\mathbf{S} : \mathbf{T} \equiv \sum_{a,b=x,y,z} S^{ab} T^{ba}. \quad (24)$$

If the system is composed of a set of species α , η can be expressed as [cf. Eq. (3)]

$$\eta = \sum_\alpha \left(\eta_{\alpha\alpha} + 2 \sum_{\beta > \alpha} \eta_{\alpha\beta} \right), \quad (25)$$

where

$$\eta_{\alpha\alpha} \equiv \frac{VY_{\alpha\alpha}(\Pi)}{6kT}, \quad (26)$$

$$\eta_{\alpha\beta} \equiv \frac{VY_{\alpha\beta}(\Pi)}{6kT}. \quad (27)$$

In terms of nonequilibrium thermodynamics, the viscous phenomenon can be interpreted as a process in which the shear stress, $-\Pi$, in Eq. (21) adiabatically provokes the velocity field, $\nabla \mathbf{v}$, which is a dyad defined in Eq. (20) between $\nabla \equiv (\partial/\partial x, \partial/\partial y, \partial/\partial z)$ and the mass center velocity of the system $\mathbf{v}$. This consequently results in internal entropy production. If we put aside the rotational viscosity, which is related to the asymmetric part of the pressure tensor, the shear viscosity η corresponds to the proportionality constant between $-\Pi$ and the traceless and symmetric parts of $\nabla \mathbf{v}$,¹⁷

$$-\Pi = 2\eta(\nabla \mathbf{v})_{\circ}^{\text{sym}} \quad (28)$$

$$= 2\sum_{\alpha} \left(\eta_{\alpha\alpha} + 2\sum_{\beta>\alpha} \eta_{\alpha\beta} \right) (\nabla \mathbf{v})_{\circ}^{\text{sym}} \quad \therefore \text{Eq. (25)}, \quad (29)$$

where

$$(\nabla \mathbf{v})_{\circ}^{\text{sym}} \equiv \frac{1}{2}(\nabla \mathbf{v} + \widetilde{\nabla \mathbf{v}}) - \frac{1}{3}(\nabla \cdot \mathbf{v})\mathbf{U}, \quad (30)$$

where $\tilde{A}$ is the transpose of A . The system flows in the direction of the shear stress $\text{sign}(-\Pi^{ab}) = \text{sign}(\partial v^b/\partial a + \partial v^a/\partial b)$, which is the velocity field in the ab plane induced by the stress in the ab plane, where $a, b = x, y, z$ ($a \neq b$). Hence, Eq. (28) perforce suggests $\eta > 0$. The heat produced per unit time $T\dot{s}$, where $\dot{s} \equiv ds/dt$ is the entropy production rate attributed to shear deformation, is given by

$$T\dot{s} = -\Pi : (\nabla \mathbf{v})_{\circ}^{\text{sym}} \quad (31)$$

$$= 2\eta \|(\nabla \mathbf{v})_{\circ}^{\text{sym}}\|^2 \quad (32)$$

$$= 2\sum_{\alpha} \left(\eta_{\alpha\alpha} + 2\sum_{\beta>\alpha} \eta_{\alpha\beta} \right) \|(\nabla \mathbf{v})_{\circ}^{\text{sym}}\|^2, \quad (33)$$

where $\|\mathbf{T}\|^2 \equiv \mathbf{T} : \mathbf{T}$. Because $\eta > 0$, $T\dot{s} > 0$, i.e., the adiabatic work inflicted on the system by shear deformation dissipates as positive heat.

III. MOLECULAR DYNAMICS SIMULATION

A. Force field

1. Yukawa potential

We adopted the potential parameters for LiPF_6 dissolved in an EC/DMC binary solvent from the previous study by Tenney and Cygan⁸ (referred to as the “TC force field”). In their study, the intermolecular forces were modeled by Coulombic interactions and the Lennard-Jones (LJ) potentials between atoms without considering spontaneous polarization. We confirmed that the TC potential without modification yields a significantly smaller diffusion coefficient and ionic conductivity than the experimentally observed values, e.g., the simulation yields only 10%–20% of the experimental conductivity. Tenney and Cygan inferred that the absence of spontaneous polarization in each species would have led to overestimating the correlated ion movement, which would have in turn underestimated the diffusivity and conductivity. The polarization effect is often introduced in MD simulations using the charge equilibration,¹⁸ dynamic fluctuating charge,¹⁹ or Drude models;²⁰ however, this often entails a heavy computational cost. For example, introducing the Drude model in the present electrolyte systems drastically

increased the total number of interactions that must be calculated, and simultaneously, it required as little as a tenth of the time step of the simulation without the Drude model to run the simulation stably. Hence, finishing the simulation within a practical time is difficult to satisfactorily integrate the time-correlation functions, which is a formidable computational task compared to static properties, such as the structure and density.

In the field of plasma physics, screened Coulombic interactions are often modeled by the Yukawa potential (also referred to as the Debye–Hückel potential) $e^{-\kappa r}/r$, where κ^{-1} represents the screening length.^{21–25} Although the underlying premise in these studies is considerably different from that in the present work, the computational cost can be considerably reduced if we succeed in introducing the screening effect by simply substituting the Yukawa potential for the usual Coulombic potential, i.e., by replacing

$$\frac{q_i q_j}{r_{ij}} \rightarrow \frac{q_i q_j}{r_{ij}} \exp(-\kappa r_{ij}) \quad (34)$$

in Eq. (1) of Ref. 8 in the TC force field, where r_{ij} is the distance between the point charges q_i and q_j . Here, we treat κ as an empirical parameter to reproduce the experimentally observed diffusion coefficients of DMC, EC, Li^+ , and PF_6^- in the EC/DMC binary solvent and its 1 mol kg^{-1} LiPF_6 solution (see Sec. IV A for details). The present study demonstrated that the Yukawa potential successfully reproduced the experimentally observed transport properties (diffusion coefficient, ionic conductivity, and shear viscosity) with minor modifications to the TC force field.

2. Other modifications to the TC force field

a. Scaling the parameter d in the LJ potential. Because the original TC force field underestimated the system density, we uniformly scaled the parameter d_{ij} in the LJ potential between atoms i and j ,

$$E_{\text{LJ}}(\epsilon_{ij}; d_{ij}; r_{ij}) = 4\epsilon_{ij} \left\{ \left(\frac{d_{ij}}{r_{ij}} \right)^{12} - \left(\frac{d_{ij}}{r_{ij}} \right)^6 \right\}, \quad (35)$$

by 0.983 for all combinations of i and j from the original TC parameters. Consequently, as shown in Fig. S6(a), the simulation reproduces the experimentally observed density for both the solvent and solution under the conditions of $p/\text{atm} = 1.0$ and $T/\text{K} = 300$. In the MD simulation, we observed that the system density considerably influences the transport properties, e.g., the diffusion coefficient. Therefore, reproducing the experimental density is significant for dealing with these properties.

b. Modifying the atomic charges in PF_6^- . We reduced the charge on F in PF_6^- from the original TC charge ($-0.361|e|$) to $-0.317|e|$ (that on P is $+0.902|e|$, the balance to maintain the formal charge of PF_6^- to $-1.0|e|$). With the original TC charge, the interaction between Li^+ and PF_6^- appears to be too strong, even after introducing the Yukawa potential to satisfactorily reproduce their diffusion coefficients.

c. Modification in the dihedral potential of DMC. To reproduce the experimentally observed fraction of DMC conformers, we modified the intramolecular dihedral potential of DMC as follows: The potential associated with the dihedral angle ϕ at $\text{O} = \text{C}-\text{O}-\text{C}$

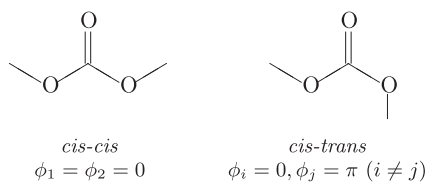


FIG. 1. Conformers of DMC; ϕ_i ($i = 1, 2$): dihedral angle at O = C–O–C.

of DMC that determines the stability of the *cis-cis* and *cis-trans* conformers (Fig. 1) is given by

$$E_n(K, d; \phi) = K\{1 + \cos(n\phi - d)\} \quad (36)$$

with $n = 1$. The parameters in the original TC force field, $K/\text{kJ mol}^{-1} = 5.86$ and $d = \pi$ were replaced in this study by $K/\text{kJ mol}^{-1} = 0.67$ and $d = 0$ (see Sec. IV B 1 for details). The dihedral potential that destabilizes the gauche conformation E_2 in Eq. (36) was unchanged from the original TC force field.

All other parameters, such as intramolecular interactions (bonds, angles, dihedrals, and impropers) and energetic parameters of the LJ potential ϵ_{ij} in Eq. (35), were adopted without changing the TC potential.

B. Conditions of the MD simulation

1. General description

MD simulations were performed using the Large-scale Atomic/Molecular Massively Parallel Simulator (LAMMPS) package.²⁶ Owing to the simple force field adopted in this study, the simulation proceeded successfully with a single time step of $\Delta t/\text{fs} = 1.0$. The simulation under the NpT condition proceeded with $p/\text{atm} = 1.0$ and $T/\text{K} = 300$. When necessary, the MD was switched to the NVE condition after equilibration for more than 1 ns (10^6 steps) under NpT conditions.

As in our previous experimental study,² the solvent composition in this study is also expressed in terms of the molar fraction of EC in the solvent,

$$x_{\text{EC}} \equiv \frac{N_{\text{EC}}}{N_{\text{DMC}} + N_{\text{EC}}}, \quad (37)$$

where N_{dmc} and N_{ec} are the number of DMC and EC molecules, respectively. A binary solvent without a solute was simulated using $N_{\text{dmc}} + N_{\text{ec}} = 1250$. The solution was simulated with $N_{\text{dmc}} + N_{\text{ec}} = 1120$ and $N_{\text{Li}} = N_{\text{PF}_6} = 100$, which translates into a molality $m/\text{mol kg}^{-1} = 1$. [The constant solvent:solute ratio $(N_{\text{dmc}} + N_{\text{ec}}) : N_{\text{Li}} : N_{\text{PF}_6} = 11.2 : 1 : 1$ yields a constant molality of the solution because of the similar molecular weights of EC (88.1 g mol^{-1}) and DMC (90.1 g mol^{-1}).] The solvent composition was varied in the range $0 \leq x_{\text{EC}} \leq 0.7$ for the solvent without a solute and its LiPF_6 solution. These conditions matched those adopted in our previous experimental study.²

Most simulated physical quantities, such as the conductivity and diffusion coefficients, are presented as the mean and twice the standard error ($\pm 2\sigma_e$, 95% confidential interval) of 12–24 MD runs, for each of which different initial atomic coordinates and velocities were provided by the equilibration process described in the

supplementary material to attain the equilibrium conformer fraction of the DMC.

2. Diffusion coefficient

Diffusion coefficients D_+ , D_- , D_{dmc} , and D_{ec} were calculated from the slope of the mean-square displacement [Eq. (10)] in the range $0.5 \leq t/\text{ns} \leq 1.0$. Extending this range to $4.0 \leq t/\text{ns} \leq 8.0$ barely affected the results.

3. Ionic conductivity

Equation (12) involves atoms belonging not only to the ionic species ($z_\alpha \neq 0$) but also to neutral species ($z_\alpha = 0$), where the constituent atoms have a nonzero charge ($q_i \neq 0$). We confirmed, however, that the conductivity calculated from the correlation between all atoms in the system coincides with that calculated from only the correlation between charged species; this suggests that correlations involving neutral species practically cancel each other. In this study, conductivity is hence calculated only from the correlation between the atoms in Li^+ and PF_6^- , which significantly reduce the computational cost.

First, we calculated $Y_{\pm\pm}(q\mathbf{v})$ and $Y_{+-}(q\mathbf{v})$ using Eqs. (4) and (5), respectively, with the upper limit of the time integral $\tau/\text{ps} = 150$, where the integral converges, as shown in Fig. S1. For each run, 5×10^6 data points corresponding to 5 ns were included in the statistical average $\langle \dots \rangle$. The autocorrelation terms, $\sigma_\pm$ (i.e., the Nernst–Einstein conductivity), were calculated from the simulated diffusion coefficient $D_\pm$ using Eq. (15), from which cross terms within the cations and anions ($\sigma'_{\pm\pm}$) were calculated from $Y_{\alpha\alpha}$ using Eq. (17). The cross term between the cation and anion (σ_{+-}) was calculated from Y_{+-} using Eq. (18). The mean of 24 runs is presented.

4. Shear viscosity

Unlike the ionic conductivity, not only ions but also neutral solvents (DMC and EC) are involved in calculating the viscosity in Eq. (23). With the upper limit of the time integral $\tau/\text{ps} = 600$, $Y_{\alpha\alpha}$ (VII) and $Y_{\alpha\beta}$ (VII), where Π is the traceless part of the pressure tensor defined in Eq. (21), were calculated using Eqs. (4) and (5), respectively. The time integral converges, as shown in Fig. S2. To avoid numerical error accumulation and reduce computational load, data points were collected at every ten steps (i.e., every 10 fs) to calculate the statistical average, $\langle \dots \rangle$, during a total 6×10^6 steps (i.e., 6 ns) of the MD run. The mean of 24 runs is presented.

IV. RESULTS AND DISCUSSION

A. Determination of κ in the Yukawa potential

The screening length κ^{-1} of the Yukawa potential in Eq. (34) was determined as an empirical parameter to reproduce the experimentally observed diffusion coefficients D_α . As described in the supplementary material, the diffusion coefficient of each species α was simulated as a function of κ , $D_\alpha(\kappa)$, for a given solvent composition x_{EC} . By plotting κ^2 on the y axis and $\{D_\alpha(\kappa)/D_\alpha^{\text{exp}}\} - 1$ (where D_α^{exp} is the experimentally observed diffusion coefficient of α) along the x axis (Fig. S9), the y -intercept gives κ^2 that on average reproduces D_α^{exp} where each species is equally weighed.

Figure 2 shows optimized κ^2 for $0.1 \leq x_{\text{EC}} \leq 0.7$. For either the solvent or the solution, κ^2 monotonically decreases with x_{EC} with a

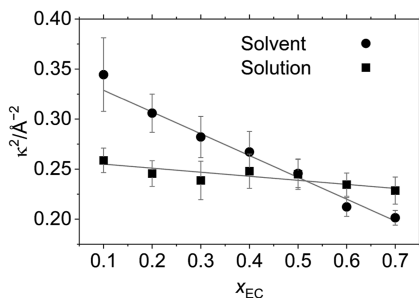


FIG. 2. The optimized screening length κ^{-1} that reproduces the experimental diffusion coefficients D_{α}^{exp} (α = DMC, EC, Li^+ , and PF_6^-) in our previous study.² To retain the equal weight of all species when optimizing κ , $x_{\text{EC}} = 0$ was excluded. (solvent) EC/DMC binary solvent without solute, and (solution) 1 mol kg^{-1} LiPF_6 dissolved in EC/DMC. The data points and error bars are the mean and twice the standard error ($\pm 2\sigma_e$) of twelve runs at each x_{EC} , respectively. The lines indicate the results of linear optimization.

greater rate of decrease for the former. Throughout this study, linearly optimized values indicated by the lines in Fig. 2 were adopted as κ values for MD simulations at each x_{EC} .

It remains as an open question how to theoretically underpin the empirically determined $\kappa^2/\text{\AA}^{-2} \approx 0.25$ or the screening length $\kappa^{-1}/\text{\AA} \approx 2$. However, one can make the following considerations. The linearized Poisson–Boltzmann (LPB) equation suggests that the screening effect leads the Coulombic interaction $\propto 1/r$ to the form $\propto \exp(-\kappa_D r)/r$. The parameter κ_D^{-1} , referred to as the Debye screening length, is expressed as¹⁶

$$\kappa_D^2 = \frac{e^2}{\epsilon_0 \epsilon_r kT} \sum_i n_i z_i^2, \quad (38)$$

where ϵ_0 , ϵ_r , and n_i are the permittivity of vacuum, relative permittivity (or dielectric constant) of the matrix, and number density of ions i , respectively. Although the LPB equation assumes the continuous nature of the matrix around ions and, hence, Eq. (38) applies only to dilute electrolyte solutions, if we tentatively estimate κ for the present system, 1 mol kg^{-1} LiPF_6 in EC/DMC, we find that κ could be a few ångströms. Equation (38) suggests that the higher the dielectric constant ϵ_r , the smaller the κ^2 value. Figure 2 is qualitatively consistent with this trend, in that (i) the dielectric constant of the binary solvent increases from $\epsilon_r = 3$ at $x_{\text{EC}} = 0$ to $\epsilon_r = 90$ at $x_{\text{EC}} = 1$, which can decrease κ^2 with x_{EC} (see also Fig. S10), and (ii) when salt is added, ϵ_r jumps up in pure DMC, whereas it drops in EC,²⁷ which leads to a less steep slope in the solution than in the solvent. A theoretical argument by Kjellander *et al.*^{28–34} demonstrated that the interaction between charges can be indeed represented by the Yukawa potential with its screening length κ^{-1} even shorter than that estimated from the LPB equation at a high salt concentration. Yet, they pointed out that κ bifurcates from a single real number into a complex number at a certain threshold, i.e., in addition to the exponentially decaying component, an oscillatory component is overlaid in the interaction, of which the effect is not considered in the present study.

B. Static properties

1. Fraction of the DMC conformers

The presence of different conformers in linear carbonates plays an important role in determining the Li^+ solvation state, and hence, it influences the transport properties.^{2,4,27,35} The *cis-trans* conformer of DMC (DMCct) is experimentally distinguishable from its *cis-cis* counterpart (DMCcc) using vibrational spectroscopy, which is also distinguishable in the MD simulation in the present study, as shown in Fig. S3(a). In the EC/DMC binary solvent without a solute at $T/K = 300$, the Raman spectra suggest that 3%–6% of the DMC is present as DMCct, and in the 1 mol kg^{-1} LiPF_6 solution, the DMCct fraction increases to 7%–10% (the *trans-trans* conformer is not observed in the solvent and solution).² From this observation, we had estimated that DMCcc is more stable than DMCct, as $\Delta E_{\text{ct-cc}}/\text{kJ mol}^{-1} = 7\text{--}8.5$, which roughly coincides with that experimentally determined by Katon and Cohen,³⁶ $\Delta E_{\text{ct-cc}}/\text{kJ mol}^{-1} = 10.9 \pm 2.1$ for pure DMC (i.e., $x_{\text{EC}} = 0$). In the isolated gas phase,^{11,37,38} the energy difference is theoretically calculated as $\Delta E_{\text{ct-cc}}/\text{kJ mol}^{-1} = 12\text{--}13$, which rationalizes the original TC dihedral potential E_1 in Eq. (36), where $\Delta E_{\text{ct-cc}} = 2 \text{ K} = 11.72 \text{ kJ mol}^{-1}$. However, we found that DMC existed exclusively in the *cis-cis* conformer in either the solvent or solution by running the MD simulation with this original dihedral potential regardless of the solvent composition, and neither the *cis-trans* nor the *trans-trans* conformer occurred. Only with a dihedral potential that slightly stabilizes the *trans* configuration did the simulation roughly reproduce the experimentally observed composition of the DMC conformers, as shown in Fig. S3(b) (cf. Sec. III A 2). The simulation also reproduced the higher fraction of the *cis-trans* conformer in the solution than in the solvent, which is due to the stabilization of the *cis-trans* conformer by its strong interaction with Li^+ . Although the intramolecular potential adopted in this study does not prohibit the *trans-trans* conformer, it never occurs during the MD simulation, which supports the experimental observations. The stability of DMCcc over DMCct is not explicitly expressed by the mechanical dihedral potential E_1 in Eq. (36). As discussed in the previous studies,^{2,10,11,39} the stability of DMC conformers appears to be governed by the inextricably intertwined relationship between charges and induced dipole moments among species present in the system; this was not explicitly modeled in this study.

2. Coordination and solvation numbers

For descriptive purposes, we distinguish between the “solvation number” (n^s) and the “coordination number” (n^c) around Li^+ . The former n^s signifies the number of molecules around Li^+ , which was experimentally estimated using Raman spectroscopy in our previous study.² The latter, n^c , is the number of molecules around Li^+ derived from the radial distribution function (RDF) obtained in the present MD simulation.

Figure S4 shows the Li^+ -centered RDF of the carbonyl oxygen (Oc) in C=O in DMC ($g_{\text{Li-Oc}(\text{dm})}$) and that in EC ($g_{\text{Li-Oc}(\text{ec})}$), and Fig. S5 shows that of P and F in PF_6^- ($g_{\text{Li-P}}$ and $g_{\text{Li-F}}$) at $x_{\text{EC}} = 0.5$ as examples of the RDFs. The intensity of the RDF changes with a change in x_{EC} , while the peak position does not change significantly. The positions of the first peak of Oc in DMC and EC (2.03 and 2.05 Å, respectively) are very close to 2.04 ± 0.01 Å observed in

the neutron diffraction experiment⁴⁰ for LiPF₆ in propylene carbonate, a cyclic carbonate analogous to EC, in which the salt-to-solvent molar ratio is 1:9 (cf. 1:11.2 in this work). They are slightly more distant than the previous MD simulations of 1.98 Å for LiPF₆ in EC/DMC.¹¹ In $g_{\text{Li-P}}$, the strong first peak indicates the presence of PF₆[−] strongly interacting with Li⁺ and that the position of the weak second peak coincides with the mean separation between adjacent Li⁺ and PF₆[−], $\bar{r}_{+-}$ —the distance between neighboring Li⁺ and PF₆[−] when they are homogeneously dispersed—calculated from the bulk number density of ions, $n_{\pm}^0$ [see Figs. S6(b) and S7 for details]. The average position of the two peaks in $g_{\text{Li-F}}$ weighed with their peak heights is ~ 2.6 Å, which is shorter yet comparable to 2.98 Å observed by the neutron diffraction.⁴⁰

The coordination number of α (=DMC, EC, PF₆[−]) around Li⁺, n_{α}^c , was calculated by integrating $g_{\text{Li-A}(\alpha)}(r)$ as

$$n_{\alpha}^c = 4\pi n_{\alpha}^0 \int_0^{r_c} r^2 g_{\text{Li-A}(\alpha)}(r) dr, \quad (39)$$

where A = Oc or P, n_{α}^0 is the bulk number density of α , and r_c represents the local minimum of $g(r)$ after the peak (see Figs. S4 and S5 and Table S I). No discernible x_{EC} dependence was observed for the peak positions or r_c . When the first peak in $g_{\text{Li-F}}$ is integrated using Eq. (39) to r_{c1} in Fig. S5, its ratio to $P n_{\text{F}}^{\text{cl}}/n_{\text{P}}^c$ ranges 1.6–1.8, implying that the PF₆[−] octahedron tends to coordinate to Li⁺ on average with its edge. The coordination is far from static, however, because the fluorine atoms within a PF₆[−] anion are exchanged hundreds of times during the residence time in the Li⁺ coordination shell, as suggested by a previous MD study.¹¹ By integrating $g_{\text{Li-F}}$ to include the second peak in r_{c2} in Fig. S5, we obtain $n_{\text{F}}^{\text{cl}}/n_{\text{P}}^c = 6$ –7, which is consistent with formula PF₆[−].

Figure 3 compares the coordination number n^c derived from the RDF with the solvation number n^s estimated by Raman spectroscopy in our previous study.² The total coordination number of three species $n_{\text{tot}}^c = n_{\text{dmc}}^c + n_{\text{ec}}^c + n_{\text{PF}_6}^c$ slightly increases with an increase from 4.8 at $x_{\text{EC}} = 0$ to 5.6 at $x_{\text{EC}} = 0.7$. The increase rate $n_{\text{tot}}^c(0.7)/n_{\text{tot}}^c(0) = 1.16$ coincides with that of the solution bulk number density, $n_{\text{tot}}^0(0.7)/n_{\text{tot}}^0(0) = 1.15$ (see the supplementary material), which reflects the invariant r_c in Eq. (39) with x_{EC} (see the supplementary material). If we count the coordination number of the solvent only, $n_{\text{solv}}^c \equiv n_{\text{dmc}}^c + n_{\text{ec}}^c$, it increases with x_{EC} from 3.5 to 5.2, which is steeper than the total n_{tot}^c , which reflects the steep decrease in $n_{\text{PF}_6}^c$ from 1.3 to 0.4. This is consistent with the increase in the degree of ionic dissociation with x_{EC} , which is experimentally concluded in our previous study.² This is also corroborated by the trend in the microscopic current correlation between Li⁺ and PF₆[−], which contributes less negatively to the ionic conductivity with an increase in x_{EC} as revealed in Sec. IV C 2 in the present study.

For given x_{EC} , the coordination number in Fig. 3(a) of the solvent is always greater than the corresponding solvation number in Fig. 3(b), $n_{\text{solv}}^c(x_{\text{EC}}) > n_{\text{solv}}^s(x_{\text{EC}})$, which suggests that the shift in the vibrational frequency of the solvent molecules caused by solvating Li⁺ reflects a smaller spatial region than that covered by the local minimum r_c in the RDF. For 1 mol dm^{−3} LiPF₆ in EC/DMC, using ¹⁷O NMR, Bogle *et al.*⁴¹ determined the number of DMC and EC molecules around Li⁺, which we refer to as “Bogle’s number,” n^{B} . Bogle’s number of solvent $n_{\text{solv}}^{\text{B}} \equiv n_{\text{dmc}}^{\text{B}} + n_{\text{ec}}^{\text{B}}$ ranges from 8 to 9 in

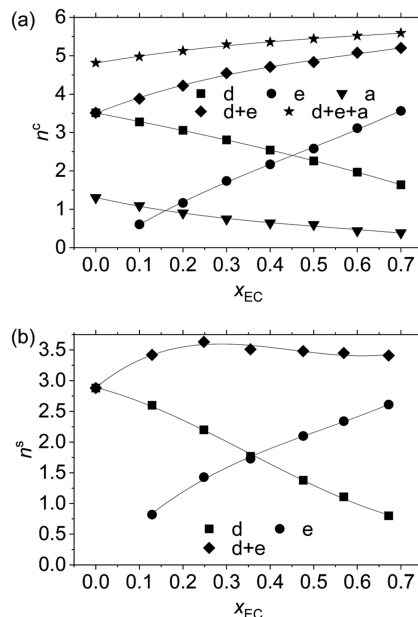


FIG. 3. (a) Coordination number, n^c , around Li⁺, calculated from the radial distribution function (RDF) of the carbonyl oxygen (Oc) in the solvent molecules (EC and DMC) and P in PF₆[−] using Eq. (39). The means of six MD runs are presented. (b) Solvation number n^s of the DMC and EC around Li⁺ estimated by the Raman spectroscopy in our previous study,² where the corresponding number for the anion is unattainable because the Raman signal from PF₆[−] is insensitive to its surroundings. Letters “d,” “e,” and “a” stand for the DMC, EC, and the anion, PF₆[−], respectively. The lines are visual guides.

the solvent composition covered in this work ($0 \leq x_{\text{EC}} \leq 0.7$), above which $n_{\text{solv}}^{\text{B}}$ decreases to 5.7 toward $x_{\text{EC}} = 1.0$. As stated by Bogle *et al.*, the spatial region probed by ¹⁷O NMR may involve both the first- and second-solvation shells, which explains the observation $n_{\text{solv}}^{\text{B}} > n_{\text{solv}}^c$. Consequently, n^s reflects the phenomena in the vicinity closest to Li⁺, followed by n^c , and n^{B} covers the most distant region.

The approximate size of the coordinated entity around Li⁺, referred to as the “Li⁺-coordination shell,” can be estimated from the coordination number n_{α}^c of each species α and its van der Waals size. Using the scheme proposed by Bondi⁴² and later modified by Zhao *et al.*,⁴³ the van der Waals volumes V^{vdw} were calculated to be 84.3, 71.9, and 68.6 Å³ for DMC, EC, and PF₆[−], respectively.⁴ The radius of the Li⁺ coordination shell is approximated as

$$\frac{4\pi}{3}(r^c)^3 = n_{\text{dmc}}^c V_{\text{dmc}}^{\text{vdw}} + n_{\text{ec}}^c V_{\text{ec}}^{\text{vdw}} + n_{\text{PF}_6}^c V_{\text{PF}_6}^{\text{vdw}}, \quad (40)$$

where the minute volume of the central Li⁺ (1 – 2 Å³) is neglected. Figure S8 compares r^c with the hydrodynamic radius (Stokes radius) r_{St} of Li⁺ derived in our previous study² from the experimentally measured viscosity and diffusion coefficients of Li⁺. While r^c is almost invariant at ~ 4.5 Å, r_{St} is significantly greater than r^c at a low x_{EC} . The gap between r_{St} and r^c shrinks with x_{EC} and $r_{\text{St}} \approx r^c$ at $x_{\text{EC}} > 0.5$. The difference $r^c - r_{\text{St}} \lesssim 0$ resembles the trend of $2\sigma_{+-} < 0$, which is the contribution from the dynamic correlation between Li⁺ and PF₆[−] to the ionic conductivity discussed in Sec. IV C 2. That

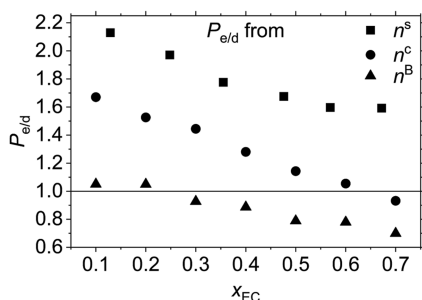


FIG. 4. The EC preference over DMC around Li^+ , $P_{e/d}$, defined in Eq. (41), which is derived from the solvation number n^s , coordination number n^c , and Bogle's number n^B .

is, $2\sigma_{+-}$ is significantly negative at low x_{EC} , but it asymptotically approaches zero at a high x_{EC} . Although its physical significance is arguable,⁴⁴ the hydrodynamic radius r_{st} measures, if properly treated, a certain dynamic effect on ionic transport phenomena wherein static indices such as r^c are not reflected.

If the EC and DMC were equally capable of solvating or coordinating Li^+ , their ratios $n_{\text{ec}}^a/n_{\text{dmc}}^a$ ($a = s, c$, or B) should be equal to their presence in the bulk solution, i.e., $x_{\text{EC}}/(1 - x_{\text{EC}})$. Hence, the index

$$P_{e/d} \equiv \frac{n_{\text{ec}}}{n_{\text{dmc}}} \cdot \frac{1 - x_{\text{EC}}}{x_{\text{EC}}} \quad (41)$$

measures the EC preference over DMC.² Figure 4 compares the three sets of $P_{e/d}$ derived from the solvation number n^s , coordination number n^c , and Bogle's number n^B . The EC preference was most conspicuous in the vicinity of Li^+ , as reflected by $P_{e/d}^s > 1$ at any solvent composition. Conversely, in terms of Bogle's number, EC is less preferred than DMC ($P_{e/d}^B < 1$) for most solvent compositions. The trend $P_{e/d}^s > P_{e/d}^c > P_{e/d}^B$ is consistent with the conclusion drawn by Bogle *et al.*⁴¹ that, while EC overwhelms DMC in strongly interacting with Li^+ in its vicinity, the presence of DMC is prevalent, albeit not prevailing, when assessed in a wider environment around Li^+ . In terms of n^s , n^c , or n^B , $P_{e/d}$ decreases with x_{EC} . Behind this phenomenon lies a balance between counter-acting effects, in that the Li^+ -solvent attraction is somewhat countered by the repulsion between solvating (or coordinating) solvent molecules, whose dipole moments are forced to repulsively align by the central Li^+ .¹⁰ The greater dipole moment of EC compared to that of DMC renders this competition more pronounced for the former.

C. Transport properties

1. Diffusion coefficient

Figure 5 compares the simulated diffusion coefficients of each species D_α in (a) a binary solvent without a solute and (b) its 1 mol kg^{-1} LiPF_6 solutions with those experimentally determined using the pulse-gradient spin-echo NMR.² In either solvent or solution, an increase in x_{EC} decreases the diffusivity of all species. It is not surprising that D_α values on average coincided with the experiment because we fixed the parameter κ in the manner described in Sec. IV A. However, for a given x_{EC} , the determined κ common to

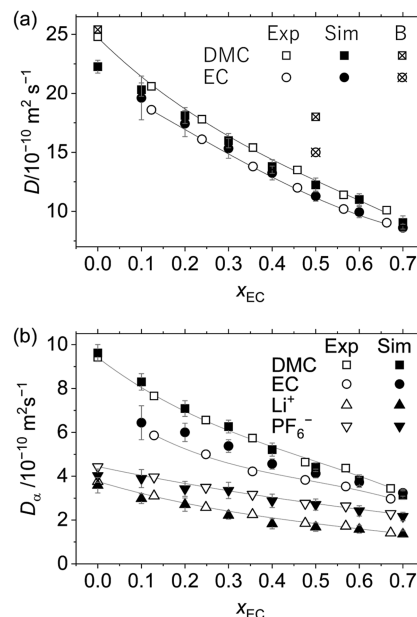


FIG. 5. Diffusion coefficient D_α in the (a) EC/DMC binary solvent without a solute and (b) 1 mol kg^{-1} LiPF_6 solution at $T/K = 300$. (Exp) Experimentally determined by pulse-gradient spin-echo NMR² and (Sim) simulated results, where the points and error bars indicate the mean and twice the standard error ($\pm 2\sigma_\theta$) of the mean of 12 runs for each x_{EC} . (B) MD simulations by Borodin and Smith.¹¹

all species (i.e., independent of each species) successfully reproduces D_α for each species. The simulation is consistent with the experiment in the following two features: (i) When LiPF_6 is dissolved in the binary solvent, both D_{dmc} and D_{ec} drop to approximately one-third, but the latter decreases more than the former at low x_{EC} and the gap between them narrows at high x_{EC} . (ii) In the solution, the gap between D_- and D_+ widens with x_{EC} . In our experimental study,² we concluded that these observations confirm the greater capability of EC in solvating Li^+ , which enhances the dissociation between Li^+ and PF_6^- and leads to a higher ionic conductivity, yet the preferred solvation of EC to Li^+ over DMC wanes with an increase in x_{EC} . This experimentally inferred mechanism of the transport phenomena in terms of ionic dissociation is corroborated in the present study from the perspective of correlations between ions, as discussed in Sec. IV C 2.

When the inter-charge interaction is modeled by the “normal” Coulombic interaction without exponentially decaying term ($e^{-\kappa r}$), we confirmed that the diffusion coefficients are significantly underestimated; we found that LiPF_6 hardly dissociated into ions at any x_{EC} . The Yukawa potential models well the screening effect in electrolyte solutions, obviating the need to explicitly model the complex polarization effect in dealing with the transport properties. One may suspect that, in either the solvent or solution, the monotonic decrease in κ (or increase in screening length κ^{-1}) with x_{EC} shown in Fig. 2 is responsible for a decrease in D_α . However, the variance in κ with x_{EC} has a minor effect on the decrease in D_α because even with a constant $\kappa/\text{\AA}^{-1} = 0.5$, a significant decrease in D_α with x_{EC} is observed, as shown in Fig. S11.

2. Conductivity

The GK conductivity [Eq. (13)] is explicitly expressed in the present system, where charge carriers consist of one cation (Li^+) and one anion (PF_6^-) as a sum of five terms,

$$\sigma = \sigma_+ + \sigma_- + \sigma'_{++} + \sigma'_{--} + 2\sigma_{+-}. \quad (42)$$

The component $\sigma_{\text{NE}} \equiv \sigma_+ + \sigma_-$ corresponds to the Nernst–Einstein conductivities of the cation and anion, which are related to their diffusion coefficients through Eq. (15); σ'_{++} and σ'_{--} account for the contribution to σ from the cross correlation between the cations and between anions in Eq. (17); and $2\sigma_{+-}$ accounts for the contribution from the cross correlation between the cation and anion in Eq. (18). Figure 6(a) shows that the sum of five terms in Eq. (42) agrees well with the experimentally observed conductivity. By introducing the Yukawa potential and fixing its screening length κ^{-1} so as to reproduce the experimental diffusion coefficients, the force field also reproduces well the conductivity. Each term in Eq. (42) is shown in Fig. 6(b). Conductivity components arising from the auto-correlation of Li^+ , σ_+ and that of PF_6^- , σ_- (i.e., the Nernst–Einstein conductivity), which were calculated from their diffusion coefficients using Eq. (15), monotonically decrease with x_{EC} because of the decrease in D_+ and D_- shown in Fig. 5(b). The other three terms negatively contribute to the conductivity, i.e., $\sigma'_{\pm\pm} < 0$ and $2\sigma_{+-} < 0$. The terms $\sigma'_{\pm\pm}$ are negative because the repulsion between ions with the same sign drives them in the opposite direction, i.e., their velocity correlation is negative ($Y'_{\pm\pm}(\mathbf{v}) < 0$), and all other terms are positive in Eq. (16). The term $2\sigma_{+-}$ is negative because the attraction between ions of opposite sign tends to move together, i.e., their

velocity correlation is positive ($Y_{+-}(\mathbf{v}) > 0$), and $z_+z_- = -1 < 0$ in Eq. (18). This scenario contrasts with that of single ionic liquids or molten salts, in which the fluid consists of only two species (a cation and an anion). In this case, the physical constraint dictates that, although counterintuitive, the unique cation–anion correlation positively contributes to the conductivity $\sigma_{+-} \geq 0$.⁴⁵ In fact, in molten alkali metal carbonates A_2CO_3 ($\text{A} = \text{Li}, \text{Na}, \text{K}, \text{Rb}, \text{and Cs}$), we previously demonstrated using MD simulations⁴⁶ that $\sigma'_{\pm\pm} < 0$, but $\sigma_{+-} > 0$ when only one alkali metal species is involved. Only in their mixtures, where a couple of cation–anion correlations are involved, do some of them contribute negatively to the conductivity. For instance, in the binary mixture $(\text{Li}_{1-x}\text{Cs}_x)_2\text{CO}_3$, while the $\text{Cs}^+-\text{CO}_3^{2-}$ correlation contributes positively to the conductivity, the $\text{Li}^+-\text{CO}_3^{2-}$ correlation significantly attenuates the conductivity, which explains the minimum in σ at $x \approx 0.5$.

Figure 6(b) indicates that $\sigma'_{\pm\pm}$ is approximately constant ($\approx -2.5 \text{ mS cm}^{-1}$), regardless of the solvent composition. In contrast, $2\sigma_{+-}$ is considerably negative in pure DMC, $2\sigma_{+-}(x_{\text{EC}} = 0)/\text{mS cm}^{-1} \approx -17$. The presence of neutral species drastically differentiates the conduction mechanism from that of the solvent-free fluids. When EC is added, $2\sigma_{+-}$ becomes less negative, crosses $\sigma'_{\pm\pm}$ at $x_{\text{EC}} \approx 0.4$, and approaches zero asymptotically. This observation suggests that the conduction mechanism switches with the solvent composition as discussed below.

Factor Δ that measures the deviation of the real conductivity σ from the Nernst–Einstein conductivity σ_{NE} is given by¹⁵

$$\sigma = \sigma_{\text{NE}}(1 - \Delta), \quad (43)$$

where $\Delta > 0$ in many cases, i.e., the correlation between ions dampens the ionic conductivity from the hypothetical scenario in which the cross correlations between ions are absent and only the auto-correlations (i.e., self-diffusion) are considered. It is occasionally considered that Δ reflects the “ion pairing,” or alternatively, $\alpha \equiv 1 - \Delta$ reflects the “ion dissociation.” In our previous experimental study,² we found that α monotonically increased from ~ 0.25 to ~ 0.65 when x_{EC} increased from 0 to ~ 0.7 , which roughly agrees with the “dynamic degree of ion dissociation, α_d ,” calculated by Borodin and Smith.¹¹ Equation (42) suggests that

$$\Delta = -\frac{\sigma'_{++} + \sigma'_{--} + 2\sigma_{+-}}{\sigma_+ + \sigma_-}. \quad (44)$$

This relation indicates that Δ reflects not only the dissociation between ions with opposite signs ($2\sigma_{+-}$) but also the repulsion between ions with identical signs ($\sigma'_{\pm\pm}$). Figure S12 plots the proportion f_{+-} of $2\sigma_{+-}$ in the total negative effect on conductivity,

$$f_{+-} \equiv \frac{2\sigma_{+-}}{\sigma'_{++} + \sigma'_{--} + 2\sigma_{+-}}. \quad (45)$$

In pure DMC ($x_{\text{EC}} = 0$), the cation–anion correlation accounts for $\sim 80\%$ of the negative effect. However, with an increase in x_{EC} , f_{+-} decreased considerably, and at $x_{\text{EC}} \gtrsim 0.5$, it decreased to only $\sim 10\%$. Thus, although at low x_{EC} the mechanism that impedes ionic conduction may be interpreted as poor ionic dissociation, it is predominantly the cation–cation and anion–anion repulsions that reduce the conductivity at a high x_{EC} . The “degree of ion dissociation α ” derived from Eq. (43) underestimates the real nature of dissociation. Considering that $f_{+-} \approx 0.1$ at $x_{\text{EC}} \gtrsim 0.5$ (or that $2\sigma_{+-} \rightarrow -0$

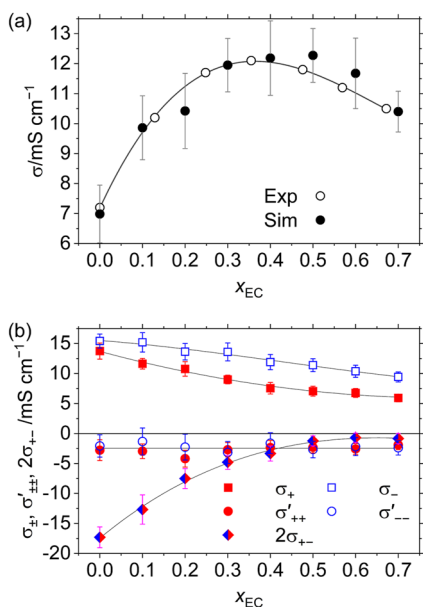


FIG. 6. (a) Specific conductivity σ of $1 \text{ mol kg}^{-1} \text{ LiPF}_6$ in the EC/DMC binary solvent at $T/K = 300$. (Exp) Experimentally measured in our previous study² and (Sim) simulated in this study. (b) Breakdown of σ into five components in Eq. (42). The lines are visual guides. Error bars indicate twice the standard error of the mean of a dozen MD runs.

at high x_{EC} , EC's ability to dissociate Li^+ and PF_6^- is hardly fulfilled above this EC content. Adding more EC provokes the adverse effect of viscous EC that impedes ionic movement, as reflected in the decrease in D_+ and D_- , which consequently levels off σ at $x_{EC} = 0.4$ – 0.5 , as shown in Fig. 6(a).

We demonstrated that a characteristic feature of the ionic conductivity of electrolyte solutions could be modeled using a mechanism in which ions of the opposite sign and neutral species come close to each other, but the strong repulsion between ions of the identical sign prevents them from being placed at the nearest-neighbor distance.^{47–49} The present molecular dynamics simulation is in line with this mechanism, i.e., (i) $2\sigma_{+-}$ is significantly influenced by x_{EC} because it is related to the phenomena taking place in the close vicinity of ions (e.g., EC's dominance over DMC in coordinating Li^+ to expel PF_6^- out of the vicinity of Li^+ as indicated by the coordination number n^c discussed in Sec. IV B 2) and (ii) $\sigma'_{\pm\pm}$ are little influenced by x_{EC} because they are related to the phenomena taking place in a more distant region around ions (e.g., the region measured by Bogle's number n^B , in which the dominance of EC fades out as discussed in Sec. IV B 2).

Because the electrolyte in the present system, $LiPF_6$, is a single-component and anti-symmetric $z_+ = -z_- = 1$, one can conceptually divide the net conductivity σ into two: the contributions from the cation and the anion, by postulating that they equally share the cross correlation term $2\sigma_{+-}$ in Eq. (42), viz., Eq. (42) can be written as

$$\sigma = \tilde{\sigma}_+ + \tilde{\sigma}_-, \quad (46)$$

where

$$\tilde{\sigma}_{\pm} \equiv \sigma_{\pm} + \sigma'_{\pm\pm} + \sigma_{+-}. \quad (47)$$

Hence, the indices

$$\theta_{\pm} \equiv \frac{\tilde{\sigma}_{\pm}}{\sigma} \quad (48)$$

account for the cation and anion contributions to the net conductivity ($\theta_+ + \theta_- = 1$). Figure S13 shows that the cation accounts for ~25%–35% of the net conductivity, which roughly coincides with the cation transport number, t_+ , experimentally measured for nonaqueous lithium electrolyte systems similar to the present study.^{50–52} The transport number is occasionally described as a one-particle property, e.g., in terms of the mobility of each ionic species, μ_{α} . Yet, if we are to equate the experimentally observed transport number $t_{\pm}$ with $\theta_{\pm}$ in Eq. (48), concepts such as μ_{α} not only represent one-particle properties but also implicitly involve correlations with other particles.

3. Viscosity

Figure 7 compares the experimentally observed shear viscosity of the EC/DMC binary solvent without a solute and its 1 mol kg⁻¹ $LiPF_6$ solution with that calculated by integrating the correlation function of the off-diagonal terms of the pressure tensor, Eq. (25). The MD simulation reproduced well the viscosity of both the solvent and solution, although at high x_{EC} it somewhat underestimated the experimental results.

For a binary solvent without a solute, Eq. (25) is explicitly written as a sum of the following three terms:

$$\eta = \eta_{dd} + \eta_{ee} + 2\eta_{de}, \quad (49)$$

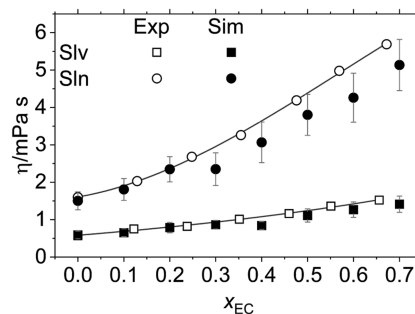


FIG. 7. Shear viscosity η of the EC/DMC binary solvent without the solute and its 1 mol kg⁻¹ $LiPF_6$ solution at $T/K = 300$. (Exp) Experimental,² where the lines are visual guides, and (Sim) calculated using Eq. (23), where the error bars span twice the standard error of the 24 MD runs.

where “d” and “e” stand for DMC and EC, respectively (Fig. 8). The terms η_{dd} and η_{ee} are contributions to η from DMC and EC, respectively, which are derived from the correlation between DMC and EC [cf. Eq. (4)], and the term $2\eta_{de}$ is derived from the correlation between DMC and EC; cf. Eq. (5). Because the number of DMC and EC molecules linearly decreases and increases, respectively, with an increase in x_{EC} , η_{dd} decreases and η_{ee} increases. The rate of increase in η_{ee} is greater than that of the decrease in η_{dd} . The correlation between DMC and EC, $2\eta_{de}$, increases at low x_{EC} and levels off or decreases at high x_{EC} . Each term in Eq. (25) can be viewed in terms of “per molecule” by defining η^m as

$$\eta_{\alpha\alpha}^m \equiv \frac{1}{\xi_{\alpha}} \eta_{\alpha\alpha} = \frac{N_{tot}}{N_{\alpha}} \eta_{\alpha\alpha}, \quad (50)$$

$$\eta_{\alpha\beta}^m \equiv \frac{1}{\sqrt{\xi_{\alpha}\xi_{\beta}}} \eta_{\alpha\beta} = \frac{N_{tot}}{\sqrt{N_{\alpha}N_{\beta}}} \eta_{\alpha\beta}, \quad (51)$$

where $\xi_{\alpha} \equiv N_{\alpha}/N_{tot}$ is the mole fraction of α in the entire system (in a solvent without a solute, $\xi_e = x_{EC}$ and $\xi_d = 1 - x_{EC}$). Figure S14(a) shows plots of the per-molecule components η_{dd}^m , η_{ee}^m , and $2\eta_{de}^m$. Despite the higher viscosity of EC than DMC in its pure state (cf. Table I), the per-molecule viscosity $\eta_{ee}^m < \eta_{dd}^m$ at low x_{EC} . With an increase in x_{EC} , while η_{dd}^m is given, the error, almost constant (or only slightly increases), η_{ee}^m significantly increases to surpass η_{dd}^m at the equimolar mixture, $x_{EC} = 0.5$. The per-molecule correlation between DMC and EC, $2\eta_{de}^m$, also increases with x_{EC} , yet less significant than η_{ee}^m . Equation (29) implies that $-\Pi_{\alpha\beta} \equiv 2\eta_{\alpha\beta}(\nabla\mathbf{v})_{\alpha\beta}^{sym}$ can be viewed as shear stress on α arising from the velocity field of β and vice versa ($\Pi_{\alpha\beta} = \Pi_{\beta\alpha}$). The relative invariance of η_{dd}^m with x_{EC} suggests that the per-molecule shear stress on DMC to flow itself, $-\Pi_{dd}/\xi_d = -\Pi_{dd}/(1 - x_{EC})$, is not significantly affected by the presence of the EC. In contrast, a greater stress on each EC molecule, $-\Pi_{ee}/\xi_e = -\Pi_{ee}/x_{EC}$, is required to provide the same velocity field to EC when EC is surrounded more by itself. The more flexible nature of the linear carbonate DMC than that of cyclic EC may be related to these observations.

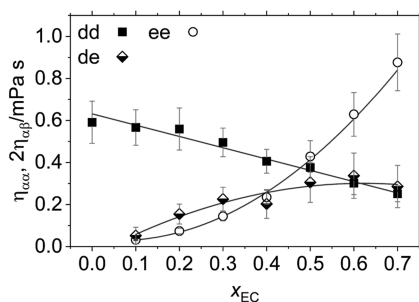


FIG. 8. Breakdown of viscosity η into the components in Eq. (49), where “d” and “e” stand for DMC and EC, respectively. The error bars span twice the standard error of the 24 MD simulations; the lines are visual guides.

For the solution, LiPF_6 dissolved in EC/DMC, Eq. (25) becomes a sum of following ten terms:

$$\eta = \eta_{dd} + \eta_{ee} + \eta_{++} + \eta_{--} + 2(\eta_{de} + \eta_{d+} + \eta_{d-} + \eta_{e+} + \eta_{e-} + \eta_{+-}), \quad (52)$$

where subscripts d, e, +, and – stand for DMC, EC, Li^+ , and PF_6^- , respectively. The net simulated viscosity of the LiPF_6 solution, shown in Fig. 7, is divided into components in Fig. 9.

In terms of the per-molecule viscosity, η^m defined in Eqs. (50) and (51), $\eta_{dd}^m \approx \eta_{ee}^m$, and both steadily increase with an increase in x_{EC} in the solution. Unlike in the solvent without solute, where the increase in η is mainly attributed to the presence of EC, both DMC and EC nearly equally contribute to the increase in η in the solution [compare Figs. S14(a) and S14(b)]. The anion contributes more to the net viscosity than the cation, $\eta_{--} > \eta_{++}$. The increase in x_{EC} influences η_{++} little, but it significantly decreases η_{--} and η_{+-} . This phenomenon seems to be related to the role of EC in liberating PF_6^- from the interacting with Li^+ while maintaining Li^+ being heavily and constantly solvated by the solvent, as discussed in Secs. IV B 2 and IV C 2. This is also in line with what was experimentally explored previously,² where the decrease in the rate of the hydrodynamic Stokes radius, $-(1/r_{St})(d r_{St}/d x_{EC})$, is the greatest for PF_6^- . Both η_{++} and η_{--} contributed less to the net viscosity than DMC and EC. However, by considering their molar ratio ($N_{dmc} + N_{ec} = 11.2N_{+}$), the per-molecule viscosity is the greatest for the anion, followed by the cation, i.e., $\eta_{--}^m > \eta_{++}^m > \eta_{dd}^m \approx \eta_{ee}^m$.

The most salient feature that distinguishes the solution from the solvent in Fig. 8 is that, while in the solvent all the terms are positive, negative cross terms, $\eta_{\alpha\beta}$, appear in the solution. Among the six cross terms, $\eta_{\alpha\beta} (\alpha \neq \beta)$, only the correlation between Li^+ and PF_6^- is positive, i.e., $\eta_{+-} > 0$. The other five terms, all of which involve the solvent species, are negative or close to zero, i.e., $\eta_{d\pm}, \eta_{e\pm}, \eta_{de} \lesssim 0$. Equations (29) and (33) suggest that the shear stress arising from these correlations points opposite to the net stress and velocity field, $\text{sign}(-\Pi_{\alpha\beta}^{ab}) = -\text{sign}(-\Pi^{ab}) = -\text{sign}(\partial v^b/\partial a + \partial v^a/\partial b)$, and hence, it negatively contributes to heat dissipation. The correlation between DMC and EC, η_{de} , and its slope, $d\eta_{de}/dx_{EC}$, changed from positive in the solvent to negative in the solution. The net increase in η with x_{EC} is attributed to

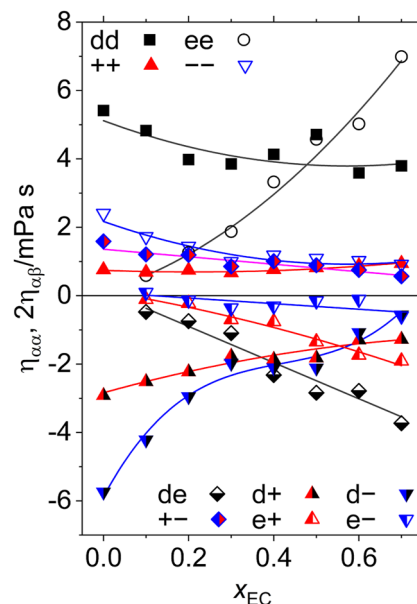


FIG. 9. Breakdown of viscosity η into components in Eq. (52), where “d,” “e,” “+,” and “–” stand for DMC, EC, Li^+ , and PF_6^- , respectively. The error bars are omitted for the visual clarity. The lines are visual guides.

the significant increase in $\eta_{dd} + \eta_{ee}$ and $2\eta_{d\pm}$. These components overwhelm the terms η_{--} , $2\eta_{de}$, and $2\eta_{e\pm}$, which decrease with x_{EC} .

V. CONCLUSIONS

The MD simulation successfully reproduced the experimentally observed transport properties of the 1 mol kg^{-1} LiPF_6 electrolyte dissolved in the binary solvent EC/DMC as a function of the solvent composition x_{EC} , i.e., the diffusion coefficient (Fig. 5), ionic conductivity (Fig. 6), and shear viscosity (Fig. 7). The simulated abundance ratio of DMC conformers roughly coincided with that determined by spectroscopy, as described in Sec. IV B 1.

Conductivity and viscosity were analyzed in terms of dynamic correlations between the constituent species. The cation–anion cross correlation plays a key role in determining the solvent composition dependence of conductivity, which further substantiates the notion that EC facilitates ionic dissociation, as discussed in our previous experimental study² and those cited therein. However, above a certain EC content, the effect of dissociation is limited because of the asymptotic approach of the cation–anion correlation to zero. At a high EC content, it is repulsive correlations between ions with identical signs that dominantly dampen the real conductivity from the hypothetical conductivity estimated from the diffusion coefficient (the Nernst–Einstein conductivity). Unlike the phenomena related to the cation–anion attractive interaction, those associated with the repulsion between ions are often experimentally unobservable, and therefore, it can occasionally be precarious to analyze the conduction mechanism only in terms of simple concepts, such as solvation, ion dissociation, or ion pairing.

For the viscosity, all three correlations are positive when the solute is absent. In contrast, in the presence of the electrolyte, cross

correlations involving solvent species are negative, which implies that these terms negatively contribute to the entropy production associated with shear deformation, which is always net positive. Unlike conductivity, where velocity correlations can be related to articulate indices (e.g., the cation–anion correlation to their dissociation), linking correlations of the pressure tensor to certain tangible concepts is not straightforward, and this left the present discussion somewhat abstract. Further theoretical conceptualization is a future task.

The introduction of the Yukawa potential as an interchange interaction—admittedly, a debatable scheme—considerably reduced the computational cost, which allowed us to directly integrate the correlation functions to precisely calculate the transport properties. Although the screening length $1/\kappa$ was treated merely as an empirical parameter, another open question that remains unanswered is the theoretical basis that justifies (or falsifies) its validity.

SUPPLEMENTARY MATERIAL

Auxiliary figures (Figs. S1–S14) and a table (Table S I) are available in the supplementary material.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Tetsu Kiyobayashi: Conceptualization (lead); Formal analysis (equal); Investigation (equal); Writing – original draft (lead). **Satoshi Uchida:** Investigation (equal); Writing – review & editing (equal). **Hirofumi Ozaki:** Data curation (equal); Software (equal). **Kenji Kiyohara:** Data curation (equal); Formal analysis (equal); Software (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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